Sai Srivatsa Ravindranath

CONTACT INFORMATION
http://saisrivatsa.com/
saisrivatsan12@gmail.com

INTERESTS

Foundation models for Recommendation, Ranking, and Retrieval
Researching and developing foundation models for recommender systems, spanning novel architectures, semantic IDs, constrained decoding, task-specific post-training & RL, and latency-aware inference. Focused on building steerable, scalable recommendation systems that optimize user engagement and business objectives.

Multi-agent systems, Mechanism Design, Algorithmic Economics
Developing deep learning and reinforcement learning methods for economic design problems involving incentives, strategic behavior, and interactions among agents.

EDUCATION

Harvard University
Ph.D in Computer Science
Advisor: Prof. David Parkes

Indian Institute of Technology, Kharagpur
B. Tech (with Honors) in EE, Minor in CS

WORK EXPERIENCE

Spotify Research (Sept 2025 - Present)
Research Scientist,Personalization
- Foundation models for large-scale recommendation, ranking, and retrieval

Google Research (July 2024 - Feb 2025)
Student Researcher, OMEGA
- Speculative decoding for large language model inference through the lens of economic theory, mechanism design, and cost-quality-latency trade-offs.

Google Research (June 2023 - Jan 2024)
Student Researcher, Market Algorithms
- Deep reinforcement learning method for sequential combinatorial auctions

Microsoft Research (2016 - 2017)
Research Fellow, Machine Learning and Optimization
- Scalable tree and embedding-based methods for multi-label learning over very large output spaces, with applications to ranking and recommendation.

JOURNAL ARTICLES

$^{\alpha}$ *denotes alphabetical ordering of authors*

Data Market Design through Deep Learning
SS. Ravindranath*, Y. Jiang*, DC. Parkes
- Forthcoming in Mathematics of Operations Research, 2026
- Also appeared at NeurIPS 2023

ArXiv: https://arxiv.org/pdf/2310.20096.pdf

Automated Mechanism Design: A Survey
MJ. Curry, Z. Fan, Y. Jiang, SS. Ravindranath, T. Wang, DC. Parkes
SIGecom Exchange, 2025
ArXiv: https://www.sigecom.org/exchanges/volume_22/2/CURRY.pdf

Optimal Auctions through Deep Learning: Advances in Differential Economics$^{\alpha}$
P. Dutting, Z. Feng, H. Narasimhan, DC. Parkes, SS. Ravindranath
- Journal of the ACM (JACM), September 2023
- Also features in Communications of the ACM Research Highlights
- Preliminary version appeared at ICML'19

ArXiv: https://arxiv.org/pdf/1706.03459.pdf

Preprints

Strategic Foundation Models
D. Goktas, A. Greenwald, T. Osogami,. . ., SS. Ravindranath et. al.
Position Paper, 2025

Deep Reinforcement Learning for Sequential Combinatorial Auctions
SS. Ravindranath, Z. Feng, D. Wang, M. Zaheer, A. Mehta, DC. Parkes
ArXiv: https://arxiv.org/abs/2407.08022

Conference & Workshop Papers

Who Are We Recommending To? Recommender Systems in the Agentic Web
H. Abdollahpuri, K. Kretschman, SS. Ravindranath, J. Doremus, M. Lalmas
Twentieth ACM Conference on Recommender Systems (RecSys 2026)

Framing and Signaling: An LLM-Based Approach to Information Design
S. Hossain, T. Lin, P. Duetting, RP. Leme, H. Xu, S. Zuo, SS. Ravindranath
Economics and Computation (EC 2026)

Deep Learning for Two-Sided Matching
SS. Ravindranath, Z. Feng, S. Li, J. Ma, SD. Kominers, DC. Parkes
Sixth International Workshop on Matching Under Preferences (MATCH-UP 2022)
ArXiv: https://arxiv.org/pdf/2107.03427.pdf

From Predictions to Decisions: Using Lookahead Regularization
N. Rosenfeld, S. Hilgard, SS. Ravindranath, DC. Parkes
Thirty-Fourth Conference on Neural Information Processing Systems (NeurIPS 2020)
ArXiv: https://arxiv.org/pdf/2006.11638.pdf

Salient Object Detection via Objectness Measure
SS. Ravindranath, RV. Babu
Twenty-Second International Conference on Image Processing (ICIP 2015)
ArXiv: https://arxiv.org/pdf/1506.07363.pdf

Learning Objective functions for Improved Image retrieval
SS. Ravindranath, M. Gygli, LV. Gool
MediaEval Workshops, 2015.

Book Chapters

Machine Learning for Matching Markets$^{\alpha}$
Z. Feng, DC. Parkes, SS. Ravindranath.
In F. Echenique N. Immorlica and V. Vazirani, editors
Online matching theory and market design. Cambridge University Press, 2022.

Machine Learning for Optimal Economic Design$^{\alpha}$
P. Dutting, Z. Feng, N. Golowich, H. Narasimhan, DC. Parkes, SS. Ravindranath.
In JF Laslier, H. Moulin, MR. Sanver, WS. Zwicker, editors,
The Future of Economic Design. Springer, 2019

Teaching

CS 136: Economics and Computation (Teaching Fellow)
Harvard University, Fall 2021

Advising

A.B Thesis in Applied Math/Computer Science co-advised with Prof. David Parkes.

Dominik Bohnet Zurcher (Harvard → Oxford)
- Pick Me: Reducing Wastefulness in the RSD Mechanism

Jeff (Yanchen) Jiang (Harvard → Google Research)
- Learning to Sell Information

Christopher En (Harvard → Columbia)
- Introduction to Auction Theory

Scholarships, Achievements

Inspire Fellowship for Higher Education
Program by Dept. of Science and Technology, Govt. of India

Kishore Vaigyanik Protsahan Yojna Fellowship (KVPY)
Awarded to top 250 students in India by Dept. of Science and Technology, Govt. of India

Certificate of Merit in:

- Indian National Mathematics Olympiad (INMO)
- National Standard Examinations in Chemistry (NSEC).
- National Standard Examinations in Physics (NSEP).

National Talent Search Scholarship (NTSE)
Awarded to top 1000 high school students in India by NCERT

Professional Services

Conference Reviewing
NeurIPS (2021 - Present), ICML (2023 - Present), ICLR (2023 - Present)

Journal Reviewing
Mathematics of Operations Research

Workshop Reviewing
AAAI 2025 Workshop on Markets, Incentives, and Gen AI
ICLR 2022 Workshop on Gamification and Multiagent Solutions